

Itô Lemma: $dX_t = \mu_t dt + \sigma_t dW_t$ ←

$d f(t, X_t) = ?$

$\Delta f(t, X_t)$

$= f(t_2, X_{t_2}) - f(t_1, X_{t_1}) =$

$$\underbrace{f(t_2, X_{t_2})}_{\Delta f} = \underbrace{f(t_1, X_{t_1})} + \underbrace{\frac{\partial f}{\partial t}}_{(t_1, X_{t_1})} \cdot (t_2 - t_1) + \underbrace{\frac{\partial f}{\partial X}}_{(t_1, X_{t_1})} (X_{t_2} - X_{t_1}) + \underbrace{\frac{1}{2} \frac{\partial^2 f}{\partial t \partial X}}_{(t_1, X_{t_1})} (t_2 - t_1)(X_{t_2} - X_{t_1}) + \underbrace{\frac{1}{2} \frac{\partial^2 f}{\partial t^2}}_{(t_1, X_{t_1})} (t_2 - t_1)^2 + \underbrace{\frac{1}{2} \frac{\partial^2 f}{\partial X^2}}_{(t_1, X_{t_1})} (X_{t_2} - X_{t_1})^2 + \dots$$

$0 = t_0 < t_1 < t_2 < \dots < t_n = t$

$\Delta f(t, X_t) = f(t, X_t) - f(0, X_{t_0})$

$= \sum_{k=0}^{n-1} \underbrace{f(t_{k+1}, X_{t_{k+1}}) - f(t_k, X_{t_k})}_{\Delta f}$

$= \sum_{k=0}^{n-1} \frac{\partial f}{\partial t} (\overbrace{t_{k+1} - t_k}^{\Delta t_k}) + \sum_{k=0}^{n-1} \frac{\partial f}{\partial X} (X_{t_{k+1}} - X_{t_k}) + \dots + \dots$

if $\Delta t_k \rightarrow 0$

$= \int_0^t \frac{\partial f}{\partial s} ds$

if $\Delta t_k \rightarrow 0$

$= \int_0^t \frac{\partial f}{\partial X} dX_s$

$$\underbrace{(X_{t_{k+1}} - X_{t_k})^2}_{\substack{\downarrow \\ \Delta t_k \rightarrow 0}} = (dX_t)^2 = \mu_t^2 (dt)^2 + \sigma_t^2 (dW_t)^2 + 2\mu_t \sigma_t dt dW$$

$$dX_t = \mu_t dt + \sigma_t dW_t$$

$$\sum_{k=0}^{n-1} (\Delta X_t)^2 \xrightarrow{\Delta t_k \rightarrow 0} \underbrace{\int \mu_t^2 (dt)^2}_0 + \underbrace{\int \sigma_t^2 (dW_t)^2}_\sum (\Delta W_t)^2 \sigma_t^2 + \underbrace{\int 2\mu_t \sigma_t dt dW}_0$$

$\frac{\partial^2 f}{\partial x^2} \sum (\Delta W_t)^2 \sigma_t^2 \xrightarrow{P} \mathbb{E} \{ \sum (\Delta W_t)^2 \sigma_t^2 \}$
 $\frac{\partial^2 f}{\partial x^2} \mathbb{E} \{ (\Delta W_t)^2 \} = \text{Var}(\Delta W_t) = \underline{\Delta t_k}$

$$\xrightarrow{P} \int_0^t \underbrace{\sigma_s^2}_{\frac{1}{2} \frac{\partial^2 f}{\partial x^2}} ds$$

$$\Delta f = \int_0^t \frac{\partial f}{\partial s} ds + \int_0^t \frac{\partial f}{\partial x} dX_s + \int_0^t \frac{1}{2} \frac{\partial^2 f}{\partial x^2} \sigma_s^2 ds$$

$$df = \underbrace{\frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dX_t}_{\text{Itô Specific}} + \underbrace{\frac{1}{2} \frac{\partial^2 f}{\partial x^2} \sigma_t^2 dt}_{\text{Itô Specific}}$$

□

SDE

$$dX_t = \left(u_t^{\text{target}}(X_t) + \frac{\sigma_t^2}{2} \nabla \log p_t(X_t) \right) dt$$

$$\min_{\theta} \mathbb{E} \left\{ \left\| u_t^{\theta}(\cdot) - \underbrace{u_t^{\text{target}}(\cdot)}_{u_t^{\text{target}}(\cdot|Z)} \right\|^2 \right\} + \sigma_t dW_t$$

$$\min_{\theta} \mathbb{E} \left\{ \left\| \underbrace{\nabla_x \log P_t(\cdot)}_{\nabla_x \log P_t(\cdot|z)} - s_{\theta}(t, \cdot) \right\|^2 \right\} = \mathbb{E} \left\{ \left\| \nabla_x \log P_t(\cdot|z) - s_{\theta}(t, \cdot) \right\|^2 \right\} + C$$

$$\begin{aligned} \mathbb{E} \left\{ \left\| \nabla_x \log P_t(\cdot) \right\|^2 + \left\| s_{\theta}(t, \cdot) \right\|^2 - 2 \langle s_{\theta}(t, \cdot), \nabla_x \log P_t(\cdot) \rangle \right. \\ \left. - \left\{ \left\| \nabla_x \log P_t(\cdot|z) \right\|^2 + \left\| s_{\theta}(t, \cdot) \right\|^2 - 2 \langle s_{\theta}(t, \cdot), \nabla_x \log P_t(\cdot|z) \rangle \right\} \right\} \\ \mathbb{E} \left\{ -2 \langle s_{\theta}(t, \cdot), \nabla_x (-\log P_t(\cdot) + \log P_t(\cdot|z)) \rangle \right\} \\ P_t(z|\cdot) = \frac{P_t(\cdot|z) P_{\text{data}}(z)}{P_t(\cdot)} \quad \nabla_x \log P_t(z|\cdot) \} \end{aligned}$$

$$\log P_t(z|\cdot) = \log P_t(\cdot|z) + \underbrace{\log P_{\text{data}}(z)} - \log P_t(\cdot)$$

$$\nabla \log P_t(z|\cdot) = \underbrace{\nabla \log P_t(\cdot|z)} - \nabla \log P_t(\cdot)$$

$$= \mathbb{E}_x \left\{ \widehat{\mathbb{E}}_{z|x} \left\{ -2 \langle s_{\theta}(t, x), \nabla \log P_t(z|x) \rangle \right\} \right\}$$

$$= \mathbb{E}_x \left\{ -2 \left\langle \int \nabla \log P_t(z|x) \cdot P_t(z|x) dz, s_{\theta}(t, x) \right\rangle \right\} = 0$$

$$\frac{\nabla P_t(z|x)}{P_t(z|x)} \cdot P_t(z|x)$$

$$\nabla \int \underbrace{P_t(z|x)}_1 dz = 0$$

$$p_t(x|z) = \mathcal{N}(\alpha_t z, \beta_t)$$

$$\begin{aligned} t=0 \\ \alpha_t &= 0 \\ \beta_t &= 1 \end{aligned}$$

$$\begin{aligned} t=1 \\ \alpha_t &= 1 \\ \beta_t &= 0 \end{aligned}$$

$$\log \left\{ \frac{1}{\sqrt{2\pi} \beta_t} \exp \left\{ -\frac{\|x - \alpha_t z\|^2}{2\beta_t^2} \right\} \right\}$$

$$\nabla_x \left\{ C + -\frac{\|x - \alpha_t z\|^2}{2\beta_t^2} \right\} = 0 + -\left(\frac{x - \alpha_t z}{\beta_t^2} \right)$$

① Sample $z \sim D_{\text{train}}$

② Sample $t \sim \text{Unif}(0,1)$ α_t, β_t from the schedule

③ Determine the target $-\left(\frac{x - \alpha_t z}{\beta_t^2} \right)$ $x \sim \mathcal{N}(\alpha_t z, \beta_t)$
 $x = \alpha_t z + \beta_t \epsilon$
 $\epsilon \sim \mathcal{N}(0, I)$

④ Update θ based on Grad of the loss $\|S_\theta(t, x) - \left\{ -\frac{(x - \alpha_t z)}{\beta_t^2} \right\}\|^2$

$$dX_t = \left\{ u_t^{\text{target}}(X_t) + \frac{\sigma_t^2}{2} S_\theta(t, X_t) \right\} dt + \sigma_t dW_t$$

Langevin Dynamics